

Mortal Computation Needs a Clock

Coherence–Rupture–Regeneration as a Temporal Grammar for Finite Machines

Alexander Sabine

Independent Researcher · alexander@activeinference.institute

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Abstract

In one paragraph. Mortal computation, as Hinton introduced it and as Ororbias and Friston have formalised it, holds that what an intelligent system knows is inseparable from the substrate that knows it: when the hardware dies, the knowledge dies, and the only channel across that death is distillation. The free energy principle supplies the thesis with its within-regime mechanics — a mortal computer is an open system holding a non-equilibrium steady state, allostatically regulated, self-evidencing. What the framework does not yet supply is a clock: a principled statement of when one steady state ends and the next begins; when a physical learner should stop accumulating and consolidate; what, and in what proportion, a dying system should transmit; and on what schedule plasticity should fall across a machine’s development. This paper offers Coherence–Rupture–Regeneration (CRR) for that role. Three equations and one dial, $\Omega = \varepsilon/\ell$: ℓ the geodesic extent of the system’s statistical manifold, fixed by its symmetry class up to Čencov’s normalisation constant and never fitted; ε the system’s resolution, $1/\sqrt{N}$ by Cramér–Rao, and measurable as the squared coefficient of variation of its rupture intervals. Coherence is accumulated Fisher–Rao arc length in units of ε ; a contentless rupture falls when accumulation exhausts the count the arena affords ($C \cdot \Omega = 1$); regeneration draws on the system’s own past under a maximum-entropy salience kernel $e^{C/\Omega}$ with Ω in the temperature slot. From the grammar we obtain: a per-rupture Landauer cost of at least $k_B T \ln 2$ — the bit acquired at the edge — which makes any finite accumulator a mortal computer by construction rather than by engineering choice; mortality as nested, regimes within regimes, consolidation as a small death with a small inheritance; an intrinsic consolidation trigger for task-free continual learning; an interval law for ruptures — Gamma with shape $1/\Omega$, $CV = \sqrt{\Omega}$ — that already matches the canonical bistable system with no parameter; a principled weighting for what distillation should carry, with definite content once Ω is indexed by regime; a native implementation of regeneration on thermodynamic sampling hardware (set $E = -C$, $T = \Omega$, and the archive becomes a programmable energy-based model); and a functional form, $\Omega_{\text{eff}} = \Omega/\sqrt{1-f}$, for the protected development of young physical learners scaffolded by mature ones. Predictions are stated as predictions, with falsifiers. The paper contains no new data; a companion paper develops the same grammar on the developmental and clinical side.

1 Introduction: what mortal computation establishes, and what it leaves open

Hinton’s closing argument in the forward-forward paper runs as follows. Separate software from hardware and you obtain immortality — the same program, the same knowledge, on any device — at a ruinous energy cost, because every device must be engineered to behave identically at the bit level. Let learning exploit the particular analogue properties of one physical device and you obtain efficiency at the price of mortality: the parameter values are meaningless to any other device, and the computation dies with the hardware. Two problems follow immediately, and Hinton names both. There must be a learning procedure that works on hardware whose precise details are unknown, since the device can no longer be modelled exactly from outside. And there must be a channel across death; the only one available is distillation, in which a teacher shows a student its responses and the student mimics them.

Ororbias and Friston generalise the thesis into a foundation. In their treatment a mortal computer is characterised through the Markov blanket formalism and the circular causality of inference, learning and selection; it is an open system holding a non-equilibrium steady state, regulated homeostatically and allostatically, and underwritten by the free energy principle. The framing organises a large biomimetic programme — neuromorphic hardware, in-materia computation, organoid intelligence — around a single claim: intelligent behaviour is inseparable from persistence in the face of an environment that could dissolve

it. Kleiner has since pressed the philosophical point: on the ontic reading, a mortal computation must deal with, and may exploit, properties of its own hardware that no outside description captures. Kiefer has given the thesis its thermodynamic form: a living agent is a constrained maximiser of entropy whose knowledge is an energetic constraint and whose action is an endogenous entropy flow, which is why its computation is mortal.

We take all of this as established and ask a question inside it. A non-equilibrium steady state is a description in which, on average, nothing changes: entropy is produced, flows persist, the regime holds. The framework is therefore superb on persistence within a regime and silent on succession between regimes. When does a steady state give out? When should a physical learner stop accumulating and consolidate? On what schedule should plasticity fall across a machine’s development, and when death arrives — as for a mortal computer it must — what, exactly, should cross the distillation channel, and in what proportion? These are clock questions. That the free energy principle does not answer them is not a defect; it was never meant to. It is a division of labour waiting for its other half.

Coherence–Rupture–Regeneration (CRR) is offered for that half. It is not a mechanism and adds nothing to the mechanics: Friston, Ramstead and Sakthivadivel have recently set out general machinery for modelling coupled systems in free-energy terms, and CRR alters none of it. What CRR supplies is a temporal grammar — a statement of when a way of being is used up, what falls at that moment, and what the next one is built from. Two prior instalments exist. An empirical paper at AGI 2026 established phase-gating between precision channels within the grammar (a strongly non-uniform Z_2 phase distribution at $SO(2)$ rupture, $\chi^2 = 8,041$; an optimal prior-to-sensory precision ratio of $\sqrt{2}$) and separated the grammar’s credit weighting from reinforcement-learning discounting. A companion working paper develops the same grammar through Winnicott’s transitional space, holding and the still face; the present paper is the machine-facing member of the pair. Section 2 states the grammar; Section 3 derives its mortality; Section 4 matches five current bottlenecks with five offers; Section 5 states predictions with falsifiers; Sections 6–8 locate the proposal relative to the predictive-coding programme, state what the paper does not do, and close.

2 The grammar, briefly

2.1 Three operations, in ordinary language

Anything that persists — a habit, a wound healing, a crystal forming, a learning machine — does three things in a cycle. It accumulates: it builds up structure that works, distinctions it can make, a way of going on. No finite thing accumulates forever, so at some point the current way of being is used up and it ruptures: a single moment with no duration and exactly one bit of content — it happened. Afterwards it regenerates, and the only material available is its own past; but not all of the past gets an equal vote, and how sharply the weighting discriminates is the single most consequential property of the system.

2.2 The equations

A system’s momentary state is a point on a statistical manifold: at the level of description chosen for the analysis, the distribution over the distinctions it can draw — for a system modelled under the free energy principle, the belief its internal states hold over external states, which Parr, Da Costa and Friston show to carry an information geometry for any system with a Markov blanket. The Fisher–Rao speed $L = \|\dot{x}\|_g \geq 0$ is the local rate at which it accrues distinguishable structure, and $\varepsilon = 1/\sqrt{N}$ is the smallest step it can resolve, fixed by the Cramér–Rao bound from the evidence N it integrates per distinction. Coherence is arc length in units of that step,

$$C(x, t) = \frac{1}{\varepsilon} \int_0^t L(x, \tau) d\tau, \quad (1)$$

accumulated along the system’s information-geometric trajectory and counted in the system’s own cells. Fisher–Rao distance is measured in units of distinguishability — for a Gaussian family with known σ , the distance between two means is $|\Delta\mu|/\sigma$, a z-score — so C is a pure number, and so is everything built from it. At unit resolution ($N = 1$) C is arc length itself.

When accumulated coherence reaches the count the arena affords, the regime is finished and a cut falls:

$$\delta(\text{now}) \quad \text{when} \quad C \cdot \Omega = 1, \quad \Omega = \varepsilon/\ell, \quad (2)$$

with ℓ the geodesic extent of the arena, so that a regime holds $\ell/\varepsilon = 1/\Omega$ cells. Since $C = s/\varepsilon$ with s the arc length traversed, the cut falls when $s = \ell$, after time ℓ/L at constant speed, whatever the resolution: the

period of the cycle is set by the speed and the extent; the resolution sets how many cells the period holds, and with it the breadth of memory. The Dirac delta has zero duration and carries one unit of mass; its whole information content is that it happened and nothing about what it is. Every rupture is a single polarity flip of the temporal boundary, $\Theta : 0 \rightarrow 1$ with $d\Theta/dt = \delta$ — a Z_2 event, the same for anything that ruptures at all, whatever the symmetry class of the substrate manifold. Two words attach to $C \cdot \Omega = 1$ and carry different expectations: saturation is the geometric idealisation, a bound attained exactly; exhaustion is the physical claim, a threshold crossed with modest overshoot because real evidence arrives in quanta — one spike, one batch, one turn. We use exhaustion throughout, and Section 2.4 takes the quanta literally.

After the cut the system rebuilds from its own history:

$$R(x, t) = \int \phi(x, \tau) e^{C(x, \tau)/\Omega} \Theta(t - \tau) d\tau, \quad (3)$$

where ϕ is the reconstruction resource — the field of past states available to be drawn on — and the Heaviside step admits only the settled past. This is the maximum-entropy weighting under a constraint on $\langle C \rangle$, with C in the role of minus an energy and Ω in the role of temperature. Crucially the kernel weights by how coherent a moment was, not by how recent: it is a salience kernel, not a decay kernel, and that is what makes the grammar non-Markovian at the process level. $C(x, \tau)$ in (3) is coherence as counted at the time, in the cells of the regime in which it was lived; a regime lived at high resolution carries its higher count forward.

2.3 The dial: ℓ is derived, ε is measured

By Čencov’s theorem the Fisher–Rao metric is the unique metric invariant under sufficient statistics, up to a positive constant. Once the symmetry class of the arena is fixed, therefore, its extent ℓ is fixed up to that constant, and ratios of lengths in the same metric are convention-free. The dial is

$$\Omega = \frac{\varepsilon}{\ell},$$

the fraction of the arena one resolvable step occupies. Its denominator is never fitted. Its numerator is a property of the system — the evidence it integrates per distinction — and it is measurable (Section 2.4). Capacity, ℓ/ε cells per regime, and memory breadth, the kernel’s exponent range $1/\Omega^2$, are one property, set by ε ; the rupture period ℓ/L is set by the speed. Table 1 gives the two elementary classes.

class	manifold	ℓ	Ω at $\varepsilon = 1$	CV = $\sqrt{\Omega}$	skew = $2\sqrt{\Omega}$
Z_2 (bistable)	Bernoulli arc (diameter)	π	$1/\pi$	0.564	1.13
SO(2) (rotational)	S^1 (circumference)	$2\pi\sqrt{I(\mu)}$	$1/2\pi$ at $I(\mu) = 1$	0.399	0.80

Table 1: The two elementary symmetry classes at unit resolution. The Bernoulli extent is π under $g = \sum dp_i^2/p_i$ ($\pi/2$ under the Fubini–Study normalisation). The circle’s Fisher–Rao circumference depends on the concentration of the circular family — $2\pi\sqrt{\kappa I_1(\kappa)/I_0(\kappa)}$ for von Mises — and equals 2π at unit Fisher information per radian ($\kappa \approx 1.8$); the class fixes the topology, the family fixes the length. The interval columns are the quantal law of Section 2.4; the ratio of the two CVs is $\sqrt{2}$.

Within a regime C runs from 0 to $C^* = 1/\Omega$, so the exponent C/Ω runs from 0 to $1/\Omega^2$, and the shape of the past follows it directly. Low Ω : a handful of intensely coherent past moments carry nearly all the weight, and whatever happens, the system rebuilds the same thing — habit, expertise, crystal structure, institutional memory; reliable and closed. High Ω : the whole past contributes, no episode dominates, and what gets rebuilt is not predictable from what was there before — play, exploration, convalescence, a liquid; unreliable and open. The limits bracket everything that can run the grammar at all. At $\Omega \rightarrow 0$ — an arena of unbounded extent, or a stone whose speed has fallen to zero — the threshold is never reached and nothing turns over. At $\Omega \geq 1$ a regime holds at most one resolvable state, and at $\Omega \rightarrow \infty$ nothing representable survives. Anything that learns sits strictly between.

One further quantity is load-bearing below. A system whose regime holds $1/\Omega$ cells distinguishes $1/\Omega$ steps along it, so the last distinguishable state before exhaustion sits at

$$s^* = 1 - \Omega \quad (4)$$

in the dimensionless variable $s = C \cdot \Omega$ (exactly when $1/\Omega$ is an integer; otherwise at the last full cell). At that poised edge the system is maximally determined from inside — it has accrued all the structure its regime affords but one step — and maximally undetermined from outside, since what remains is a single binary flip worth one full bit.

2.4 The quantal clock, and the measurable dial

Evidence arrives in quanta. Let each quantum advance the clock by one cell, and let quanta arrive at rate r ; then $L = \varepsilon r$, a regime ends after $1/\Omega$ quanta, and the interval between ruptures is Gamma-distributed with shape $1/\Omega$ and rate r :

$$\text{mean} = \frac{1}{\Omega r}, \quad \text{CV} = \sqrt{\Omega}, \quad \text{skewness} = 2\sqrt{\Omega},$$

with a hazard that rises and saturates. Three consequences follow. First, Ω is measurable: it is the squared coefficient of variation of a system's rupture intervals, and $1/\text{CV}^2$ is the number of cells a regime holds. Second, $\Omega = 1$ is the memoryless channel — one quantum per regime, exponential intervals, no resolvable state within a regime — so radioactive decay, shot noise and thermal emission are the $\Omega = 1$ end of the same law, and a channel that reads as Poisson is a channel with one cell. Third, the law has already been tested once for free: at $\Omega = 1/\pi$ it gives shape π , CV 0.564 and skewness 1.13, which is the measured law of dominance durations in binocular rivalry — the canonical bistable system — where gamma fits give shape 3–4, CV 0.50–0.58. A deterministic accumulator with its noise at the wall, the engine used for earlier figures in this programme, gives intervals with a CV set by the wall's noise and no skew; the quantal clock is what the grammar's own statement about quanta implies, and it is the version the predictions of Section 5 test.

3 Every finite accumulator is a mortal computer

3.1 The Landauer cost of a cut

What is erased at rupture is not the past — the past survives in ϕ — but the accumulator: coherence resets so that the next regime can begin. At a hard cut the accumulator's value is known, $C^* = 1/\Omega$, and the reset $(C^*, n) \mapsto (0, n + 1)$ is injective on the register's occupied states; by Landauer's principle, with the measurement and erasure bounds separated as by Sagawa and Ueda, it costs nothing given the count. What a cycle cannot avoid paying for is the bit it acquires at the edge (Section 2.3) — and under a quantal cut, the entropy of the overshoot besides. So each rupture costs at least

$$E_{\min} = k_B T \ln 2, \tag{5}$$

about 3×10^{-21} J at body temperature, $k_B T \ln m$ on an arena of m outcomes; and a system maintaining a finite archive in steady state pays a further $k_B T \ln 2$ per bit displaced, up to $1/\Omega$ per regime, so that rigid systems, which make many distinctions per regime, pay more per regime than open ones. The magnitude constrains nothing about chips or nervous systems and we do not pretend otherwise; the structure is what matters. Fields, Goldstein and Sandved-Smith have made the thermodynamic cost of active inference explicit at the level of belief updating; the cost here is the cost of ending a regime, and it is additional to theirs. Any system running this grammar is dissipative by construction, its ruptures thermodynamically irreversible, its history written into the world in a way that cannot be unwritten. This is Ororbia and Friston's mortal computation, reached from the accumulator side rather than the substrate side: a CRR system cannot be copied off and cannot be run backwards, not because someone chose analogue hardware but because a finite accumulator acquires and must clear one bit per cycle. Logically reversible computation avoids the fee by uncomputing rather than clearing, which is to have no cut; a cut is the point at which uncomputation is refused. The price of having a clock is dissipation. Mortality and temporality are bought together or not at all.

3.2 Mortality is nested

In the original framing, and in most commentary since, the mortal computer is coextensive with the lifespan of its hardware: one life, one death, one distillation. The grammar refines this. Regimes nest — micro-ruptures at the timescale of attention, consolidation cycles at the timescale of sleep, developmental transitions, and death — and each level runs the same three operations with its own ε and its own Ω . A lifespan is the outermost regime. Every consolidation is a small death with a small inheritance: an accumulator cleared, a Landauer fee paid, a reconstruction weighted by the kernel. This gives the transfer problem recursive structure (Section 4.3): a system is distilling into its own successor regimes continuously, and hardware death is only the limiting case in which the student is another body.

3.3 The clock is denominated in dissipation

Fisher–Rao arc length is thermodynamic length in the sense of Salamon and Berry, Crooks, and Sivak and Crooks: the excess dissipation of a finite-time protocol is bounded below by the squared thermodynamic length divided by the duration, $\langle W_{\text{ex}} \rangle \geq \mathcal{L}^2/\Delta t$, and for a system with a single relaxation time τ the friction metric is $k_B T \tau$ times the Fisher metric, so $\langle W_{\text{ex}} \rangle \geq k_B T \tau (\varepsilon C)^2/\Delta t$. The identification requires the system’s state space to be a family of equilibrium ensembles; where it holds, C is not a metaphorical odometer but the quantity that already prices finite-time driving in stochastic thermodynamics, and $C \cdot \Omega = 1$ reads as a budget statement: a regime ends when the traverse its extent affords has been spent. Rolandi and colleagues’ energy–time–accuracy trade-offs for thermodynamic computing are the same bound seen from the hardware. For a field whose entire pitch is computing near $k_B T$, a clock denominated in the currency it spends is at minimum a convenient one.

4 Five bottlenecks, five offers

Table 2 summarises; the subsections argue. Throughout, the grammar’s posture is the one stated in the introduction: it schedules and weights; it does not replace anyone’s mechanics.

bottleneck	current practice	the grammar’s offer
learning on drifting, device-specific substrates	device-aware and robust training; periodic retraining when drift invalidates parameters	treat drift as part of the substrate’s own L ; a volatile substrate turns over more often instead of silently corrupting; regeneration rebuilds structure, not parameter values
when to consolidate (stability–plasticity; task-free continual learning)	imposed schedules: after each task, every N steps, when a validation metric sags	intrinsic trigger at exhaustion, $C \cdot \Omega = 1$, on the learner’s own information geometry with steps below ε excluded; no oracle required
what to transmit across death	distillation of input–output behaviour, weighted by recency or not at all	transmit ϕ under $e^{C/\Omega}$: the closing cells of each regime, the highest-resolution regimes preferred; recursive across nested regimes
thermodynamic hardware ahead of its workloads	map existing networks onto samplers	regeneration is native: $E = -C$, $T = \Omega$; the archive is a programmable energy-based model
no theory of machine development	curriculum heuristics	annealing as ε falling; scaffolding as $\Omega_{\text{eff}} = \Omega/\sqrt{1-f}$; graduated withdrawal

Table 2: Five bottlenecks in the mortal computation programme and what the temporal grammar offers each. Predictions and falsifiers are in Section 5.

4.1 Drift is part of the substrate’s own L

The physical-neural-network literature is admirably frank about its central difficulty. In-silico training suffers model–reality mismatch; in-situ training produces device-specific models that do not transfer across fabrication variation; and post-deployment perturbations — thermal drift, misalignment, conductance drift — invalidate trained parameters and force retraining (Wright and colleagues demonstrated the hybrid route; Momeni and colleagues’ review and Xu and colleagues’ sharpness-aware work state the standing problems plainly). Devices are meanwhile pushed close to their own noise floor to save power, which worsens everything. The standing posture is adversarial: drift is an enemy to be suppressed, modelled or retrained away.

The grammar’s reframe: a drifting substrate is a substrate with nonzero Fisher–Rao speed of its own. The speed of the superposed motion is at most $L_{\text{task}} + L_{\text{substrate}}$, with equality when the two are aligned, and three consequences follow. First, the clock compensates automatically — rupture frequency goes as L/ℓ , so a volatile substrate turns over more often instead of silently corrupting; the failure is converted from an invisible one into a scheduled one. Second, what regeneration rebuilds is coherent structure weighted by the kernel, not literal parameter values, so each rebuild is a re-fit to the device as it now is; the device’s idiosyncrasy is not an error term but the ground the next regime stands on, which is the mortal computation thesis taken seriously as an operating principle rather than a lament. Third, the rheological calibration in the companion paper applies: via the Deborah number, ordinary materials span twenty-two orders of magnitude

of relaxation time between water and granite and eleven orders of the dial, and a memristive array is pitch, not granite. The failure mode of current practice is deploying granite assumptions on pitch hardware. There is a kinship worth naming with sharpness-aware training, which seeks flat basins — regions where function is insensitive to parameter perturbation. That is robustness bought spatially; rupture-and-rebuild buys it temporally; the two should compose. What the grammar does not offer here is the local learning rule. That remains the business of forward-forward, equilibrium and predictive-coding methods (Section 6).

4.2 When to consolidate: the clock

The consolidation machinery exists and works. Elastic weight consolidation stiffens parameters important to old tasks, importance measured by Fisher information; replay methods rehearse; and the wake-sleep family alternates learning with offline consolidation — Golden, Delanois, Sanda and Bazhenov’s sleep phases with local Hebbian plasticity in spiking networks, Harun, Gallardo, Hayes and Kanan’s SIESTA with intermittent sleep. In every case, however, the schedule is imposed from outside: after each task, every N steps, when a metric sags. Task-free continual learning, in Aljundi, Kelchtermans and Tuytelaars’ phrase, names the honest setting — no oracle announces the boundaries — and it is precisely there that the imposed schedules have nothing to hold on to.

The offer is an intrinsic trigger: consolidate when the coherence accumulated since the last cut exhausts the regime’s count, $C \cdot \Omega = 1$, computed on the learner’s own manifold — the information geometry of its internal states parameterising beliefs about external states, which for a parametric learner is the natural-gradient metric on its parameters — with steps smaller than ε excluded, since arc length is a path functional and an unfiltered estimator would fire on optimiser noise after a fixed number of steps regardless of what was learned. No task boundary is needed because the boundary is not in the data; it is in the learner. The relation to elastic weight consolidation is pleasing rather than competitive: EWC evaluates the Fisher metric statically, asking which directions matter; the clock integrates the same metric along the trajectory, asking how far the system has travelled, and thresholds the answer. One is a map of stiffness, the other an odometer, and a system can carry both. Note also that the stability-plasticity dilemma is the dial under its neuroscience name: Takesian and Hensch’s molecular brakes are what raise precision and lower Ω ; Gopnik’s protected early search is Ω held high; Frankenhuis and Gopnik’s early-adversity findings read as premature collapse of the dial. Development, on this account, is annealing: ε falls as the evidence integrated per distinction grows, Ω falls with it, and the schedule’s floor is the substrate’s resolution while its ceiling is the arena.

4.3 What to transmit: the kernel

Hinton’s channel across death is distillation, and his own account of its limits is the right one: mortal systems cannot ship weights, only show behaviour. But distillation as practised is silent on the question a dying system actually faces — which episodes deserve the bandwidth? The grammar answers: transmit the reconstruction resource ϕ weighted by $e^{C/\Omega}$ — by how coherent a stretch of past was, not by how recent. Two things must be said precisely. Within a regime, at the system’s own resolution, adjacent cells differ in weight by $e^{1/\Omega}$, so the kernel selects the closing cells before each cut: at $\Omega = 1/\pi$ the final cell of a regime holds 0.96 of its weight. Between regimes, with Ω indexed by regime, the edge of regime k carries $e^{1/(\Omega_k \Omega_t)}$ in a reading at Ω_t : regimes lived at higher resolution are exponentially preferred, and the one of lowest Ω_k dominates a reading at low Ω_t . With a single Ω for every regime, all completed regimes tie. So the proposal is specific: a dying mortal computer should hand over the closing stretches of its highest-resolution regimes, not its recency buffer. This weighting is measurably different from temporal discounting: in the AGI 2026 study the two are essentially uncorrelated ($\text{corr}(\gamma^k, e^{C/\Omega}) = -0.182$), and the coherence weighting assigned credit twice as accurately at depth 50 with $16.9\times$ fewer updates. Nested mortality (Section 3) makes the proposal recursive: every consolidation is a distillation into one’s own successor regime, so the same kernel that schedules sleep decides what sleep replays, and hardware death is the limiting case in which the student is another body.

4.4 Where regeneration is native: thermodynamic samplers

Thermodynamic computing crossed a credibility threshold this year — scalable probabilistic hardware from Extropic published in *npj Unconventional Computing*, Whitelam’s generative thermodynamic computer in *Physical Review Letters*, Ball’s survey of the field in *Quanta* — and its own participants state the position candidly: present designs are about as capable as the small digital networks of 1990, the headline efficiency

figures are projections rather than measurements, and the hardware is ahead of its workloads. The thermodynamic sampling unit is a machine whose inputs are the parameters of an energy function and whose outputs are samples from the corresponding Boltzmann distribution.

The observation is that the regeneration integral is such a workload, natively. Set $E(\tau) = -C(x, \tau)$ and $T = \Omega$: the salience kernel $e^{C/\Omega}$ is the Boltzmann factor $e^{-E/T}$, and R is an (unnormalised) expectation of ϕ under a programmable energy-based model over the archive — the constrained maximum-entropy form Kiefer takes as the signature of mortal, endogenous computation. A thermodynamic sampler does not simulate that expectation; it enacts it, drawing high-coherence past states with physical noise, at a physical temperature which is the dial — annealing the device is the developmental schedule. The companion paper argues the grammar is native to neuromorphic substrates because dissipation is intrinsic there. The sharper claim is that it is most native to thermodynamic samplers, where the kernel is not computed but occupied. Conte and colleagues’ roadmap asked what thermodynamic computers are for; “sampling one’s own past at temperature Ω ” is a candidate answer that arrives with a developmental literature attached.

4.5 Protected development, and collectives

Offered as considerations, not engineering claims. The field has curriculum heuristics; it does not have a theory of protected development for physical learners — of why a young system can afford more exploration than its own budget appears to permit. The companion paper derives one from the clinic, and it decomposes holding into two contributions that current practice conflates. The first is provision of ϕ : material to rebuild from — demonstrations, teacher outputs, curated data — which is what distillation already supplies. The second is composition of the dial: a partner taking, in alternation, a fraction f of the distinctions a regime requires, so that the regime is the pair’s, its period is unchanged, and the learner’s own count at the cut is $(1 - f)/\Omega$ while its resolution ε is unchanged. The learner’s kernel exponent then runs to $(1 - f)/\Omega^2$ rather than $1/\Omega^2$, and its kernel is as flat as a solitary learner’s at

$$\Omega_{\text{eff}} = \frac{\Omega}{\sqrt{1 - f}}, \quad f < 1 - \Omega^2. \quad (6)$$

This is not function-copying; it is a pair-level property. A student coupled to a teacher that absorbs half the traverse runs at $\sqrt{2}$ times its intrinsic openness — a flatter kernel, broader search — than it could sustain by itself, which is Gopnik’s protected childhood as an engineering specification. Graduated withdrawal — failing at the rate the student can replace the resource — becomes a schedule with a functional form and two named failure modes: withdraw too fast and rebuilding lacks material; adapt too long and the resource stays external.

For collectives the rupture–regeneration cycle is an integrate-and-fire oscillation, so a shared present is a coupled-oscillator question rather than a metaphor: with the coupling acting on the speed, two clocks lock when their rate mismatch is below twice the coupling and fail sharply beyond it; below a critical coupling a population keeps private nows, above it a present held in common. Tissot, Levin, Buckley and Watson’s minimal-agent result is the template — two actions, stress-driven, with timing rather than magnitude as the operative variable. The failure mode has a name from the developmental laboratory: the still face. A system producing well-formed outputs at arbitrary timing is present, facing, and not participating, and the meta-analytic still-face literature (Mesman, van IJzendoorn and Bakermans-Kranenburg) describes what that does to the partner. Concretely, this predicts that manipulating response timing should move human–machine coordination more than manipulating output fidelity. And a system that mirrors with high fidelity while absorbing none of the traverse flattens nobody’s kernel; it can only sharpen what is already peaked — an echo chamber, given a formal description.

5 Predictions

Stated as predictions, with falsifiers, and with a discipline learned from biological tests of the grammar. One of those tests — an *E. coli* division-interval mapping — failed cleanly under the earlier wall-noise interval law (CV 0.159 against an observed 0.2–0.3); under the quantal law of Section 2.4 the same data read as $\Omega \approx 0.04\text{--}0.09$, a regime of 11–25 cells, which is a measurement rather than a miss. The framework survives such corrections; frozen mappings do not; the same standard applies here.

P1 (the trigger). In task-free class-incremental learning, hold the consolidation machinery fixed (replay, sleep-phase Hebbian consolidation, or EWC-style stiffening) and vary only the schedule: consolidate at

exhaustion — when Fisher–Rao arc length accumulated since the last consolidation, with steps below ε excluded, reaches the arena’s C^* — against periodic schedules matched for total consolidation budget, validation-triggered schedules, and a task-boundary oracle. Prediction: exhaustion-triggered consolidation matches or beats matched-budget periodic schedules on average accuracy and forgetting, and approaches the oracle. Falsifier: no advantage over matched-budget periodic scheduling.

P2 (the kernel). At equal buffer size and replay budget, sample replay in proportion to $e^{C/\Omega}$ against recency-weighted, uniform, and loss-based selection. The kernel concentrates replay on the closing cells of each regime and, across regimes, on those lived at highest resolution; the prediction is that those are the cells worth replaying: lower forgetting, with the gap widening as sequence length grows, since that is where the non-Markovian weighting departs from decay. Falsifier: no advantage over recency weighting at matched budget.

P3 (the signature). Build synthetic learners whose arenas have known symmetry class and known resolution — a bistable (Z_2) and a rotational ($SO(2)$) pair; synthetic construction dissolves the bridge-mapping problem that complicated the biological tests. Prediction: rupture intervals are Gamma-distributed with shape $1/\Omega$, coefficient of variation $\sqrt{\Omega}$ and skewness $2\sqrt{\Omega}$ — at unit resolution, CV 0.564 and 0.399 for the two classes, ratio $\sqrt{2}$ across matched pairs — and the Ω read from the intervals is the Ω that governs the kernel and the trigger. Falsifier: symmetric or left-skewed intervals; a CV ratio of 2, which would indicate noise at the wall rather than quantal accumulation; or an interval-read Ω that disagrees with the trigger’s.

P4 (scaffolding). Couple a student to a teacher that absorbs a measured fraction f of the traverse (for instance, of prediction-error variance). Prediction: the regime’s period is unchanged by f ; the student’s own count per regime scales as $(1 - f)$; its kernel entropy and exploration breadth track $\Omega/\sqrt{1 - f}$; graduated withdrawal outperforms abrupt withdrawal at matched total assistance. Falsifier: a period that changes with f , breadth insensitive to f , or no ordering between withdrawal schedules.

P5 (timing, secondary). In human–machine interaction, entraining response timing to the partner’s rupture rhythm moves coordination and reported comfort more than matched manipulations of output fidelity, and the transition from locked to unlocked with increasing timing mismatch is a cliff rather than a slope.

6 Relation to the predictive-coding programme

The clock presupposes local, forward-only mechanics and is agnostic about which. Neural generative coding, active predictive coding for control, and the programme surveyed by Salvatori, Mali, Buckley, Rao, Friston and Ororbia supply exactly the class of rules the grammar schedules — and predictive-coding systems are natively instrumented for it, since precision-weighted prediction errors live on the Fisher metric already. Integration is therefore architectural rather than algorithmic: a clock module estimating L online from quantities the system already exposes, counting arc length in cells with steps below ε excluded, triggering consolidation at exhaustion, and weighting replay and distillation by the kernel — beside the learning rule, not inside it. The needed engineering contribution is a cheap online estimator of arc length; accumulated Fisher information is not a substitute, because the square root sits inside the integral. An implementation in ngc-learn would be the natural first artefact, and is offered here as a collaboration.

7 What this paper does not do

It contains no new data; every argument is an interpretive mapping plus a stated prediction. The symmetry class of a wild, high-dimensional learned manifold is not established, and until it is derived any ℓ used there is provisional — which is why the synthetic arenas of P3 come first. ℓ is never fitted; ε is measured, and the same ε must serve the trigger, the kernel and the interval law: a system in which the three disagree has falsified the grammar’s claim that capacity, breadth and timing are one property. The phase-gating results of the AGI 2026 paper were obtained at unit resolution with the wall-noise engine, and should be re-examined under the interval law of Section 2.4. Online estimation of C is nontrivial at scale, for the reason given in Section 6. The Landauer magnitude constrains structure, not chips; Section 3 is an argument about what kind of thing a finite accumulator is, not an energy audit. The grammar is not a mechanism: it will not solve credit

assignment on unknown hardware, and it modifies nothing in the free energy principle and competes with nothing in it. The holding claims of Section 4.5 are imported from a developmental argument and function here as hypotheses. The still-face and echo-chamber remarks are design considerations, not results.

8 Closing

This paper began with a brief meeting at AGI 2026 and is offered as the longer half of that exchange. The mortal computation thesis says what kind of thing a finite intelligence is; the grammar proposes what time such a thing keeps, and gives the dial that sets its memory a number that can be read off its own rhythm. If the clock is wrong, P1–P4 are cheap ways to find out, and the falsifiers are stated. If it is right, the programme acquires what every mortal thing has: a schedule of small deaths, and something definite to hand on at each of them. The conversation has a lineage — the Active Inference Institute hosted a working discussion of the mortal computation paper with its authors and Chris Fields in 2024 — so the door this paper knocks on is, in that sense, already ajar.

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Use of AI. This draft was prepared by Claude (Anthropic) at the author’s request, working from the author’s CRR corpus and from a literature review conducted in conversation with the author in August 2026. Version 2 incorporates the corrections established in a line-by-line check of version 1: the dial as ε/ℓ , the quantal interval law and the measurability of Ω , the Landauer accounting at one bit per cut, the count-only scaffolding form, and the sub-resolution filter on the trigger. All theoretical arguments and interpretive claims originate in the author’s framework and remain the author’s responsibility.